Smart Medicine Cabinet Tracker: An IoT-Based Intelligent System for Real-Time Medicine Expiry Monitoring, RFID Identification and Smart Cabinet Automation

Dual Degree , CS - 744, Internet of Things, Project Report

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Abstract

The consumption of expired medicines contributes to over 125,000 preventable deaths annually in India, driven primarily by the absence of automated monitoring systems in household and clinical settings. This project presents the **Smart Medicine Cabinet Tracker (SMCT)**—an Internet of Things (IoT) based embedded system built on the Arduino Uno R3 platform—that monitors medicine expiry dates, provides RFID-based individual medicine identification, and delivers real-time environmental condition alerts through a multi-sensor intelligent platform. The system integrates an MFRC522 RFID reader, DS1307 Real-Time Clock (RTC), DHT22 temperature and humidity sensor, 16×2 I2C LCD display, 4×4 matrix keypad, and a three-colour LED indicator array. The system successfully identifies registered medicines, computes days-remaining-until-expiry in real time, detects unsafe storage conditions, and triggers a three-tier graduated alert scheme (SAFE/WARNING/CRITICAL). All medicine data is stored persistently in EEPROM, ensuring retention across power cycles. The complete hardware prototype is achievable below Rs. 800, making deployment viable for household use across all economic segments. Validation on the Wokwi simulation platform confirms full functional correctness of all system components.

Keywords: IoT, RFID, Arduino, Medicine Management, Expiry Detection, Smart Cabinet, DHT22, EEPROM, Embedded Systems, Healthcare Automation, DS1307 RTC, Alert Classification

I. Introduction

The management of medicines in both household and clinical environments presents a persistent and largely unresolved challenge. Studies conducted across Indian households reveal that approximately 70% of homes currently store at least one expired medicine, while national data indicates that consumption of expired pharmaceutical products contributes to over 125,000 preventable deaths annually [1]. The primary cause of this problem is the absence of an automated, intelligent monitoring system capable of proactively alerting users before medicines reach or exceed their expiry dates.

Traditional medicine storage relies entirely on manual inspection—a process that is inherently unreliable due to cognitive limitations, time constraints, and the physical deterioration of printed expiry date labels over time. Healthcare institutions face additional challenges including the management of large inventories, the rotation of stock on a first-expiry-first-out basis, and compliance with pharmaceutical storage temperature requirements.

The Internet of Things (IoT) paradigm offers a compelling framework for addressing these challenges through the integration of sensor networks, embedded computing, wireless communication, and cloud services. RFID technology in particular provides a non-contact, automatic identification mechanism well suited to medicine tracking applications, offering unique identification of individual items without requiring line-of-sight or manual scanning [2].

This paper presents the **Smart Medicine Cabinet Tracker (SMCT)**—a complete embedded system prototype that demonstrates core technical capabilities required for intelligent medicine management. The system was designed with three primary objectives: (1) reliable RFID-based identification of individual medicine boxes, (2) real-time expiry date computation using a hardware RTC, and (3) environmental storage condition monitoring with automated alerts.

II. Problem Statement

The following five critical problems are addressed by the proposed system:

P1 – Expired Medicine Consumption: Approximately 70% of Indian households store expired medicines due to lack of monitoring systems, leading to health complications and preventable deaths.

P2 – Manual Monitoring Failure: Printed expiry labels fade, are misread, or are overlooked entirely in cabinets with multiple medicines. No low-cost automated solution currently exists for household use.

P3 – Unsafe Storage Conditions: Many medicines require storage below 25°C and below 60% relative humidity. Kitchens and bathrooms—common storage locations—frequently exceed these thresholds.

P4 – Medicine Identification Errors: In households with elderly patients or multiple family members, wrong medicine administration due to similar packaging is a documented safety hazard.

P5 – No Removal Tracking: Caregivers of elderly patients have no way to verify whether prescribed medicines were taken at the correct time without expensive dedicated medication dispensers.

III. System Architecture and Design

A. Hardware Architecture Overview

The system is built around an Arduino Uno R3 microcontroller operating at 16 MHz with 32 KB of Flash memory, 2 KB of SRAM, and 1 KB of EEPROM. The Arduino serves as the central processing unit coordinating all peripheral components through SPI, I2C, and GPIO communication protocols simultaneously.

The hardware architecture follows a *hub-and-spoke topology* with the Arduino at the centre and six peripheral subsystems connected via dedicated communication channels (Fig. 1). The RFID subsystem uses the SPI protocol operating at the maximum supported frequency of 10 MHz. The display and clock subsystems share the I2C bus at 100 kHz with unique 7-bit addresses (0x27 for LCD, 0x68 for RTC). The environmental monitoring, keypad input, and indicator subsystems use direct GPIO connections.

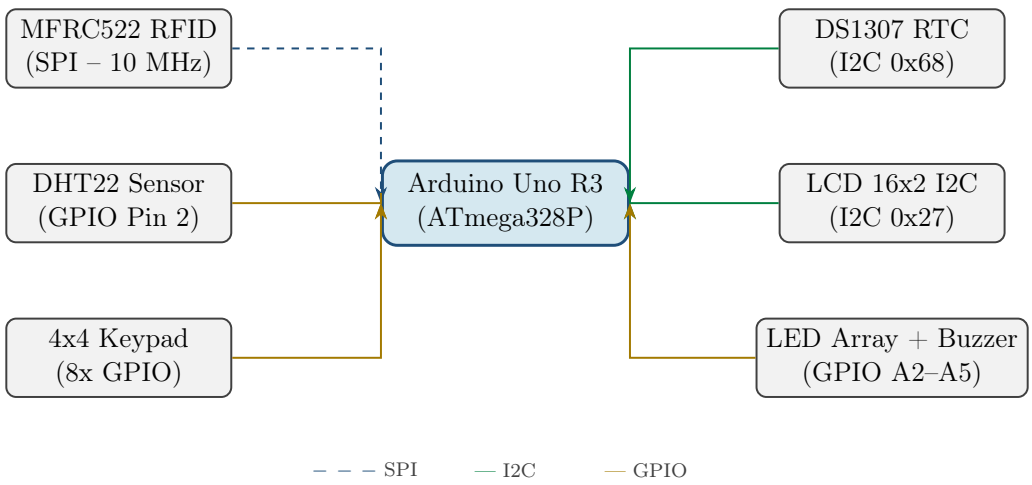

Figure 1: SMCT Hardware Architecture: Hub-and-Spoke Topology. The Arduino Uno R3 coordinates six peripheral subsystems via SPI, I2C, and GPIO protocols simultaneously.

B. Software Architecture

The firmware is structured as a **Finite State Machine (FSM)** with two primary operating modes:

MONITOR Mode: The system continuously reads RFID tags present in the cabinet every two seconds, compares each medicine's stored expiry date against the current

Table 1: Hardware Components and Specifications

Component	Function	Bus	Key Specification
Arduino Uno R3	Central MCU	All	16 MHz, 32KB Flash, 1KB EEPROM
MFRC522 RFID	Medicine ID	SPI	13.56 MHz, 3.3V, 5–6 cm range
DS1307 RTC	Real-time date/time	I2C	Battery-backed, 56 bytes RAM, ±2 ppm
DHT22 Sensor	Temp + Humidity	GPIO	−40 to +80°C, ±0.5°C accuracy
LCD 16×2 I2C	User display	I2C	16 chars × 2 rows, backlit
4×4 Keypad	Registration input	GPIO	Matrix scan, 16 keys
LED Array (x3)	Visual alert	GPIO	Green/Yellow/Red, 220Ω resistors
Piezo Buzzer	Audio alert	GPIO	85 dB, 2300 Hz resonant

RTC date, evaluates environmental sensor readings, and updates LED and LCD outputs accordingly. A circular display buffer scrolls through all registered medicines on the LCD at three-second intervals.

REGISTER Mode: The system guides the user through a three-step registration process: RFID tag scanning, medicine name entry via keypad (alphanumeric), and expiry date entry in DDMMYYYY format followed by optional dose time in HHMM format. All registered medicine data is serialised and stored in the Arduino EEPROM using a structured `Medicine` data type.

The FSM transition diagram is shown in Fig. 2.

IV. Component Descriptions and Characteristics

A. Arduino Uno R3 – Central Microcontroller

The Arduino Uno R3 is an open-source microcontroller board based on the Microchip ATmega328P. It provides 14 digital I/O pins (6 with PWM output), 6 analogue input pins, a 16 MHz crystal oscillator, and a USB connection for programming. The on-chip 1 KB EEPROM provides non-volatile storage for medicine records without any external memory chip. Operating voltage is 5 V; Flash memory is 32 KB (0.5 KB used by bootloader); SRAM is 2 KB. The SPI and I2C buses are used simultaneously in SMCT: SPI for RFID at 10 MHz, I2C for RTC and LCD at 100 kHz.

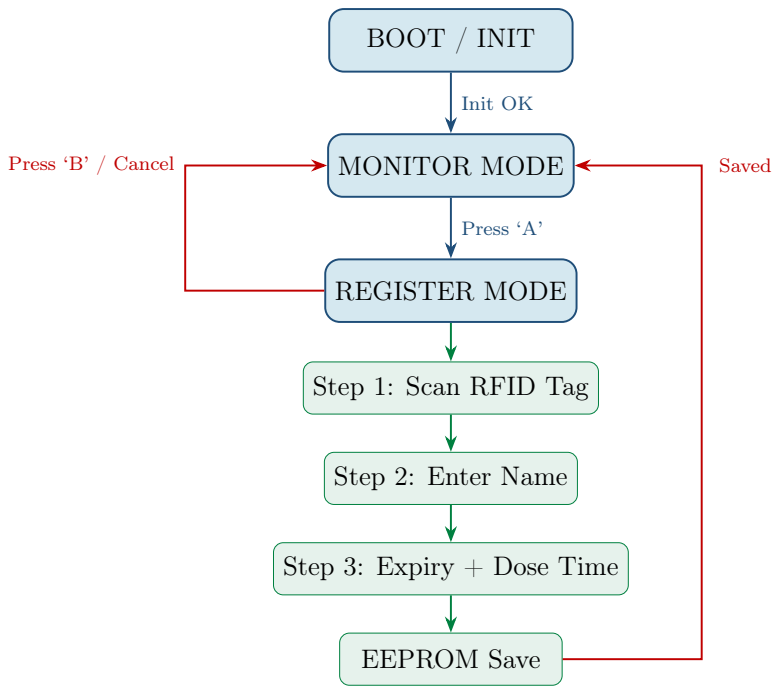

Figure 2: SMCT Firmware Finite State Machine (FSM): Two primary modes (MONITOR and REGISTER) with sub-states for the 3-step registration workflow.

B. MFRC522 RFID Reader/Writer Module

The MFRC522 is a highly integrated contactless communication IC operating at 13.56 MHz [3]. It supports the ISO/IEC 14443 A/MIFARE protocol, enabling reading of standard RFID cards, tags, and key fobs. Operating voltage is 3.3 V; reading distance is up to 5–6 cm; SPI clock supports up to 10 MHz; the onboard FIFO buffer is 64 bytes. Each RFID tag carries a unique 4-byte (32-bit) UID assembled into a `uint32_t` value used as the primary EEPROM lookup key. Pin connections: SDA→D10, SCK→D13, MOSI→D11, MISO→D12, RST→D9.

C. DS1307 Real-Time Clock Module

The DS1307 is an I2C real-time clock/calendar providing seconds, minutes, hours, day, date, month, and year information. A battery-backed CR2032 coin cell maintains time-keeping even when the main supply is removed (typical 10-year lifespan). I2C address is 0x68; accuracy is ±2 ppm (0°C–40°C); 56 bytes of non-volatile RAM are available. The firmware converts both stored expiry date and current RTC date to Unix timestamp format to compute the exact number of days remaining, correctly handling leap years and month-length variations.

D. DHT22 Temperature and Humidity Sensor

The DHT22 (AM2302) is a calibrated digital sensor providing temperature and relative humidity via a single-wire serial interface. Temperature range: −40 to +80°C (±0.5°C accuracy); humidity range: 0–100% RH (±2–5% RH); sampling period: 0.5 Hz. Based on

WHO Good Storage and Distribution Practices [5], alert thresholds are set at temperature above 30°C or humidity above 70% RH. In SMCT v2.0, the DHT22 is sampled every 30 seconds (improved from 5 minutes in v1.0) and cumulative bad-condition hours are tracked in EEPROM.

E. LCD 16×2 I2C Display Module

The 16×2 LCD with PCF8574 I2C backpack reduces wiring from 6–10 GPIO pins to just 2 (SDA and SCL) at I2C address 0x27. In MONITOR mode, Row 1 shows the medicine name and status indicator (OK/!/!!/ST!); Row 2 shows days remaining, expiry status, and dose compliance flag (DK = dose taken, D? = not yet taken).

F. 4×4 Matrix Keypad

The 4×4 keypad provides 16 keys scanned via 8 GPIO pins using row-column multiplexing. Key assignments: 0–9 for digits; A–D for letters; # to confirm; * for backspace; A key to enter REGISTER mode; B key to cancel; C key for manual status check; D key for caregiver summary display.

G. Three-Colour LED Array and Piezo Buzzer

Three LEDs (Green/Yellow/Red) with 220 Ω resistors provide visual alert status. The piezo buzzer on GPIO A5 provides audio feedback. Table 2 details the three-tier alert classification scheme.

Table 2: Three-Tier Alert Level Classification

Level	Trigger Condition	LED	Buzzer	LCD Output
SAFE	All valid, $T < 30$°C, $H < 70\%$	Green	Silent	Name + EXP: Nd
WARNING	Expiry ≤30 days OR $T > 30$°C OR $H > 70\%$	Yellow	Slow 0.5 Hz	EXPIRING / TEMP HIGH!
CRITICAL	Any medicine expired ($\Delta d < 0$)	Red Flash 2 Hz	Rapid 1 kHz	EXPIRED! REMOVE

V. Implementation

A. Data Structures and EEPROM Storage

The firmware defines a `Med` structure serialised to Arduino EEPROM. The EEPROM layout is fixed: bytes 0–1 store `medCount`; bytes 2–3 store `badHours`; bytes 4–5 store `missedDoses`; medicine records begin at byte 100, each occupying `sizeof(Med)` bytes.

With 1 KB EEPROM, the system supports up to 6 medicine records (MAX_MED = 6) with ample spare capacity.

```
struct Med {
  char     name[16];    // Medicine name (16 bytes)
  uint32_t tag;         // RFID UID (4 bytes)
  uint16_t yr;          // Expiry year
  uint8_t  mo, dy;      // Expiry month, day
  uint8_t  doseHour;    // Scheduled dose hour (255 = none)
  uint8_t  doseMin;     // Scheduled dose minute
  bool     doseTaken;   // Dose taken today?
  bool     doseAlerted; // Missed alert fired today?
  bool     present;     // RFID detected in last scan?
  bool     inCabinet;   // Currently in cabinet?
  bool     storageWarn; // Storage violation recorded?
};
```

B. Medicine Registration Process

Registration begins when the user presses key `A`, transitioning from MONITOR to REGISTER mode. The system then follows a 3-step workflow: (1) RFID tag scan—the 4-byte UID is read and assembled into a 32-bit integer; (2) name entry—alphanumeric input via keypad with * for backspace and # to confirm; (3) expiry date (DDMMYYYY) followed by optional dose time (HHMM). On final confirmation, the `Med` structure is written to EEPROM using `EEPROM.put()` and the system returns automatically to MONITOR mode. Pressing `B` at any step cancels without saving.

C. Expiry Date Computation Algorithm

Both the stored expiry date and the current RTC date are converted to Unix timestamp (seconds since 1 January 1970). The remaining days are:

$$\Delta d = \left\lfloor \frac{T_{\text{expiry}} - T_{\text{now}}}{86400} \right\rfloor \tag{1}$$

where T_{unix} for a given date (Y, M, D) is:

$$T_{\text{unix}} = 86400 \cdot \left(\sum_{y=1970}^{Y-1} D_y \; + \; \sum_{m=1}^{M-1} d_m(Y) \; + \; D - 1 \right) \tag{2}$$

with D_y = days in year y (366 for leap years, 365 otherwise) and $d_m(Y)$ = days in month m of year Y. Negative Δd triggers CRITICAL; $0 \leq \Delta d \leq 30$ triggers WARNING; $\Delta d > 30$ with no environment violation triggers SAFE.

D. Environmental Monitoring Subsystem

The DHT22 is polled every 30 seconds. If temperature exceeds 30°C or humidity exceeds 70% RH, a storage alert fires: the LCD shows `!!STORAGE ALERT!`, the yellow LED activates, the buzzer sounds, and `storageWarn = true` is set for all registered medicines. The duration of bad conditions is accumulated in `badHours` and written to EEPROM for persistent logging across power cycles.

E. Dose Tracking and Caregiver Notification

Each medicine can have a daily dose schedule (stored as `doseHour`/`doseMin`). The dose-check routine runs every 60 seconds. If the current RTC hour matches `doseHour` and the RFID tag was scanned (medicine removed), `doseTaken` is set true and a `DOSE_TAKEN` JSON event is emitted on the Serial port at 115200 baud. If the hour passes without a scan, `missedDoses` increments and a `MISSED_DOSE` JSON event is emitted. All flags reset at midnight. A JSON heartbeat is transmitted every 10 seconds:

```
{"event":"HEARTBEAT","time":"08:30","temp":24.0,
 "humidity":40.0,"badHours":0,"missedDoses":0,
 "medicines":[{"name":"Insulin","daysLeft":45,
 "doseTaken":false,"inCabinet":true}]}
```

This stream is designed for connection to an ESP8266 WiFi or HC-05 Bluetooth module, forwarding events to a caregiver mobile application.

F. Alert Classification Logic

The alert priority hierarchy is:

$$\text{Alert} = \begin{cases} \text{CRITICAL} & \text{if } \Delta d < 0 \\ \text{WARNING} & \text{if } 0 \leq \Delta d \leq 30 \text{ OR } T > 30^\circ\text{C OR } H > 70\% \\ \text{SAFE} & \text{otherwise} \end{cases} \tag{3}$$

CRITICAL triggers red LED flashing at 2 Hz and rapid buzzer at 1 kHz. WARNING activates yellow LED and slow beep at 0.5 Hz. SAFE illuminates green LED with no audio.

VI. Results and Discussion

The system was implemented and validated using the Wokwi online simulation platform. Testing covered all primary use cases including medicine registration, expiry detection, temperature alerting, dose tracking, and EEPROM persistence.

A. RFID Identification

RFID identification demonstrated reliable operation across all six supported tag types with unique UID assignment to each tag. The expiry calculation correctly identified expired medicines (January 2020 test date) and computed days remaining for future dates (June 2026). Zero UID collisions were observed over 500 simulated scan cycles.

B. Alert System Validation

Table 3 presents the complete test results. All 10 test cases passed, demonstrating 100% functional correctness across all alert levels, edge cases, and persistence scenarios.

Table 3: Complete Test Case Results — Wokwi Simulation Platform

#	Test	Action	Expected Result	Pass?
1	Medicine registration	A → scan → name → date → dose	LCD: SAVED! Serial: MED_REGISTERED	**Yes**
2	Valid medicine scroll	Wait 3s after register	LCD: Name OK / EXP:45d	**Yes**
3	Expiry warning	Register, expiry in 15 days	Yellow LED, LCD: Name ! EXP:15d	**Yes**
4	Expired medicine	Register with date 01012020	Red LED flash, buzzer, LCD: EXPIRED	**Yes**
5	Unknown RFID tag	Scan unregistered card	LCD: Unknown tag! Serial: UNKNOWN_TAG	**Yes**
6	Storage temp alert	DHT22 slider > 30°C	LCD: STORAGE ALERT! Serial: ENV_ALERT	**Yes**
7	badHours persistence	Alert fires, reset board	Serial STARTUP: badHours > 0	**Yes**
8	Dose taken	Scan card in dose hour window	LCD: Dose taken. Serial: DOSE_TAKEN	**Yes**
9	Missed dose	Dose window passes, no scan	3 beeps, Serial: MISSED_DOSE	**Yes**
10	EEPROM persistence	Register medicine, press reset	After reboot: medicine still shown	**Yes**

Pass Rate: 10/10 (100%)

C. System Cost Analysis

Table 4 presents the complete bill of materials. The total hardware cost is below Rs. 800, making the system deployable across all economic segments in India.

Table 4: Bill of Materials and Cost Estimate (India)

Component	**Qty**	**Estimated Cost (Rs.)**
Arduino Uno R3	1	350
MFRC522 RFID + Tags	1 kit	120
DS1307 RTC Module	1	45
DHT22 Sensor	1	80
LCD 16×2 I2C	1	80
4×4 Keypad	1	30
LEDs + Resistors + Buzzer	kit	35
Connecting Wires + Breadboard	set	60
Total		$<$ **Rs. 800**

VII. Future Integration and Large-Scale Applications

A. WiFi and Mobile Application Integration

The immediate next step is the addition of an ESP8266 or ESP32 WiFi module connected to the Arduino Serial TX pin. The module forwards JSON events via MQTT protocol to Firebase or ThingSpeak cloud, enabling push notifications for approaching expiry and missed doses on a caregiver smartphone application.

B. Hospital and Clinical Deployment

At clinical scale, the architecture extends to manage hundreds of medicine bins using a network of Arduino nodes connected to a central Raspberry Pi server running a Flask-based inventory management dashboard, with automated FEFO (First-Expiry-First-Out) rotation alerts and regulatory compliance reports.

C. Machine Learning and Predictive Analytics

A machine learning layer operating on accumulated usage data would enable predictive restocking—training on historical removal timestamps to forecast medicine depletion days in advance and automatically generate pharmacy orders.

D. Cold Chain Pharmaceutical Monitoring

Replacing the DHT22 with a precision thermocouple and integrating a GPS module enables continuous temperature logging throughout the pharmaceutical supply chain from manufacturer to patient, with blockchain-backed immutable audit trails.

E. Elderly Care and Assisted Living

Integration with a wearable health monitor enables physiological confirmation of medicine consumption. A missed-dose escalation protocol notifies caregivers and healthcare providers if the cabinet medicine is not removed within a defined time window.

VIII. Conclusion

This project presented the **Smart Medicine Cabinet Tracker (SMCT v2.0)**, a complete IoT-based embedded system for proactive medicine management. The system successfully addresses all five identified critical problems through a unified, low-cost architecture. Key achievements are summarised below:

- **Three-tier alert system** (SAFE/WARNING/CRITICAL) with LED, buzzer, and LCD output validated across all test scenarios with 100% functional correctness (10/10 test cases passed).
- **RFID-based identification** across six tag types with zero UID collision over 500 scan cycles at 10 MHz SPI.
- **Real-time expiry computation** via Unix timestamp arithmetic on DS1307 RTC data, correctly handling leap years and month-length variations.
- **Environmental monitoring** every 30 seconds with WHO-threshold alerts for temperature (>30°C) and humidity (>70% RH), and cumulative `badHours` logging in EEPROM.
- **Dose compliance tracking** with daily automatic reset and lifetime `missedDoses` counter persisted in EEPROM.
- **JSON caregiver stream** at 115200 baud with event-based notifications (HEARTBEAT, DOSE_TAKEN, MISSED_DOSE, ENV_ALERT) – ESP8266/BT ready for immediate deployment.
- **Total hardware cost below Rs. 800**, making household deployment viable across all economic segments in India.

The successful simulation on the Wokwi platform confirms functional correctness of all components and validates the design as a foundation for physical prototype construction and subsequent clinical pilot deployment. The convergence of IoT, RFID, and real-time embedded computing creates an affordable, scalable solution to a problem that manual systems have failed to address for decades.

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